

Charleston W.K. Chiang

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Curriculum Vitae

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Academic Appointment

2024-present	Associate Professor <i>with tenure</i> , University of Southern California Center for Genetic Epidemiology, Department of Population and Public Health Sciences Keck School of Medicine
2024-present	Associate Professor, University of Southern California (<i>joint</i>) Department of Quantitative and Computational Biology Dornsife College of Letters, Arts, and Sciences
2023-present	Associate Director, Center for Genetic Epidemiology
2022-present	Member, USC Norris Comprehensive Cancer Center Cancer Epidemiology Program
2019-2024	Assistant Professor, University of Southern California (<i>joint</i>) Department of Quantitative and Computational Biology Dornsife College of Letters, Arts, and Sciences
2018-2024	Assistant Professor, University of Southern California (<i>primary</i>) Center for Genetic Epidemiology, Department of Population and Public Health Sciences Keck School of Medicine

Education and Training

parental leaves in 2013 and 2016 (and 2022).

2017 [#]	Postdoctoral Research Fellow, UCLA Center for Neurobehavioral Genetics, Semel Institute
2015 [#]	Ruth L. Kirschstein NRSA Postdoctoral Research Fellow, UCLA Department of Ecology and Evolutionary Biology
2011	Ph.D., Genetics, Harvard University Division of Medical Sciences, Program in Biological and Biomedical Science Leder Human Biology and Translational Medicine Program Scholar Department of Genetics
2005	B.S., Microbiology, Immunology, and Molecular Genetics, UCLA Mathematics (minor), Statistics (minor) Summa Cum Laude, College Honors Program Scholar

Funded Projects and Grants

Current

Years	Role	Source	Total Costs
2021-2026	PI (PI: Chiang)	NIGMS R35GM142783	\$2,035,683
An evolutionary framework to elucidate and interpret the genetic architecture of complex traits in diverse populations			
2022-2027	PI (PI: Chiang)	NHGRI R01HG011646	\$3,996,329
Leveraging the evolutionary history to improve identification of trait-associated alleles and risk stratification models in Native Hawaiians			
2023-2027	PI (PI: Chiang)	NHGRI R01HG012605	\$2,160,123
A genome-wide genealogical framework for statistical and population genetic analysis			

2021-2026	co-I	(PI: Mancuso)	NHGRI R01HG012133	\$4,205,166
Characterizing the evolutionary architecture of complex disease within and across diverse populations				
2022-2027	co-I	(PI: de Smith)	NCI R01CA248771	\$3,815,580
Understanding the increased risk of childhood acute lymphoblastic leukemia in Latinos				
2024-2029	co-I	(PI: Darst)	NHLBI R01HL174378	\$3,587,678
Leveraging metabolomics to identify factors contributing to health disparities in Native Hawaiian individuals				
2026-2031	co-I	(PI: Shu)	NIMH R01MH143902	\$4,138,889
Characterizing autism heterogeneity by machine learning methods and its genetic and environmental drivers				

Mentored Trainee Grants

2022-2024	Mentor	(PI: Bryan L. Dinh)	NHGRI F31HG012159	\$138,108
Genomic resources and references for genetic investigation of an understudied population				
2023-2027	Mentor	(PI: Jalen Langie)	NCI F31CA278359	\$190,776
Exploring the impact of genetic ancestry on acute lymphoblastic leukemia risk in Latino populations				

Publications (Last 5 years)

A total of **32** first-, senior-, and/or corresponding- authorship (**numerically bolded**, including equal contributions) * equal contributions, † corresponding author, **bolded names** denote group member
Through 2025, 35 out of 67 publications were in journals with impact factor (IF) > 8 at the year of publication.
Through 2024, 30 out of 55 publications indexed by Clarivate's Web of Science were > 80th percentile in citations

Preprints, submitted, and in-preparation manuscripts

82. Tan Q, Xie J, Snyder LAG, Chen F, Ladd-Acosta CM, Volk H, Trost B, Fernandez B, **Chiang CWK**, Mancuso N, Shu CA. "Machine learning-derived subtypes of autism reveal distinct genetic contributions from common and rare variants." *In preparation*
81. Agasaro O*, Xu H*, Sheng X, Fuller H, Albanes D, Conti DV, Haiman CA, **Chiang CWK**, Ye K, Darst BF. "Circulating metabolite levels vary by genetic ancestry in diverse populations." *In preparation*
80. Fuller H, Guo B, Xu H, Lu Y, Haessler J, Grove M, Highland HM, Reiner A, Boerwinkle E, Yu B, Raghavan S, Albanes D, Wilkens L, Le Marchand L, Conti DV, Haiman CA, Sinnott-Armstrong N, Peters U, Kooperberg C, **Chiang CWK**, Ye K, Darst BF. "Pleiotropy in polygenic risk scores of circulating metabolite levels." *In preparation*
79. Fuller H, Guo B, Xu H, Lu Y, Haessler J, Grove M, Highland HM, Reiner A, Boerwinkle E, Yu B, Raghavan S, Albanes D, Wilkens L, Le Marchand L, Conti DV, Haiman CA, Sinnott-Armstrong N, Peters U, Kooperberg C, **Chiang CWK**, Ye K, Darst BF. "Validation of metabolite polygenic scores in Black and African American populations." *In preparation*
78. Brown SC, White K, Kana'iaupuni M, **Alatorre K**, **Dinh BL**, **Chiang CWK**, Wilkens LR, Binder A. "Genetic predisposition to type-2 diabetes increases stroke risk across diverse populations." *Submitted to Neurology*
77. Lee AJ, Neehus AL, Wahlster L, Agarwal G, Weng C, Zhang A, Liu T, **Chiang CWK**, Wiemels JL, Shelton S, Ye T, della Volpe L, Cohn O, Poeschla M, King E, Ha SA, Turvey A, de Smith AJ,

Sankaran VG. "Single-cell multiomic mapping of genetic predisposition to childhood B-cell acute lymphoblastic leukemia." *Submitted to Cancer Discovery*

76. **Dinh BL, Wang X**, Sheng X, Wan P, Srivastava AK, Naseri T, Viali S, Wilkens LR, Le Marchand L, Haiman CA, Weeks DE, **Chiang CWK**^{††}, Carlson JC^{††}. "Meta-analysis of over 8,000 individuals from Hawai'i and Samoa for genetic associations to cardiometabolic phenotypes." *In revision at PLoS Genet.*
75. Xu H, Yu G, Lu Y, Fuller H, Song S, Shen Y, **Chiang CWK**, Darst BF, Ye K. "Polygenic scores modify the associations of fish oil supplementation with circulating omega-3 fatty acids: a cross-sectional gene-diet interaction study in UK Biobank." *medRxiv* 2026: <https://www.medrxiv.org/content/10.64898/2026.04.29.26352078v1>. *In review at Clin Nutr.*
74. **Malomane DK**[†], Williams MP[†], **Tian L, Tang J**, Huber CD, Mangul S*, Abedalthagafi M^{††}, **Chiang CWK**^{††}. "Patterns of population structure and genetic variation within the Saudi Arabian population." *bioRxiv* 2026: <https://www.biorxiv.org/content/10.1101/2025.01.10.632500v2>.
73. Streicher SA, Guillermo C, Park SY, **Chiang CWK**, Sheng X, Bogumil D, Park SL, Cheng I, Stram DO, Conti DV, Haiman CA, Wilkens LR, Le Marchand L. "Characterizing cancer patterns in Okinawan vs. mainland Japanese Americans: The Multiethnic Cohort." *medRxiv* 2025: <https://www.medrxiv.org/content/10.1101/2025.07.14.25331338v1>. *In review at Int. J. of Cancer.*

Peer-Reviewed and Published

72. Li Y, Alonzo B, Zhang ZE, **Langie J**, Myint SS, Ince M, Lu R, **Chiang CWK**, Gazal S, Mancuso N, Kachuri L, de Smith AJ. "Genome-wide association study and predictors of neonatal blood cell traits in Hispanic newborns." *Am J Hum Genet.* 2026 Jul 1:S0002-9297(26)00234-X. doi: 10.1016/j.ajhg.2026.06.008. Online ahead of print.
71. Oleksyk TK, Wolfsberger WW, Chhugani K, et al. "Challenges and recommendations in establishing national human diversity genomic projects" *Nat Methods.* 2026 Jul;23(7):1261-1266. doi: 10.1038/s41592-026-03131-9.
70. **Lee EY, Dinh BL, Tang J**, Streicher SA, **Wang X, Biswas S, Tian H**, Yu X, Taparra K, Naseri T, Viali S, Weeks DE, Carlson JC, Haiman CA, Le Marchand L, Maskarinec G, Wilkens LR, Park SY, **Chiang CWK**[†]. "Incorporating dietary information to enhance polygenic prediction models with applications to body mass index and type-2 diabetes." *Genes Nutr.* 2026 Jun 18. doi: 10.1186/s12263-026-00811-1. Online ahead of print.
69. Bogumil D, Sheng X, Wan P, Xia L, Pooler L, Cheng I, Streicher SA, Huang BZ, Chen F, Stram DO, Shen J, King G, **Chiang CWK**, Ongaco CM, Adams M, McMullen I, Zhang P, Ling H, Mawhinney M, Doheny KF, Le Marchand L, Wilkens LR, Haiman CA, Conti DV. "The Multiethnic Cohort: A Resource for the study of Genetic and non-Genetic Cancer Risk Across Populations." *Cancer Epidemiol Biomarkers Prev.* 2026 May 1. doi: 10.1158/1055-9965.EPI-25-1458. Online ahead of print.
68. **Langie J**[†], **Chan TF**, Yang W, Kang AY, Morimoto L, Stram DO, Mancuso N, Ma X, Metayer C, Lupo PJ, Rabin KR, Scheurer ME, Wiemels JL, Yang JJ, de Smith AJ^{††}, **Chiang CWK**^{††}. "The impact of Indigenous American-like ancestry on risk of acute lymphoblastic leukemia in Hispanic/Latino children." *HGG Adv.* 2026 Jan 15;7(1):100534. doi: 10.1016/j.xhgg.2025.100534. Epub 2025 Oct 25.
67. Im C, Raduski AR, Mills LJ, et al. "Genome-wide association study of childhood B-cell acute

lymphoblastic leukemia reveals novel African ancestry-specific susceptibility loci." **Nat Commun.** 2025 Oct 22;16(1):8974. doi: 10.1038/s41467-025-64337-7.

66. Xu H, Sun Y, Francis M, Cheng CF, Modulla NTR, Brenna JT, **Chiang CWK**, Ye K. "Widespread but moderate genetic overlap between circulating polyunsaturated fatty acid and brain disorders." **J Lipid Res.** 2025 Oct;66(10):100890. doi: 10.1016/j.jlr.2025.100890. Epub 2025 Aug 25.
65. Wang L, Xu H, Simons RL, Beach SRH, Lei MK, Ong ML, Philibert R, Mielke MM, Sun Y, Lu Y, **Chiang CWK**, Darst BF, Ye K. "Associations of longitudinal changes in blood biomarkers of dementia with the proportion of genetically inferred African ancestry." **Neurology.** 2025 Sep 9;105(5):e213976. doi: 10.1212/WNL.0000000000213976. Epub 2025 Aug 8.
64. Lu Y, Xu H, Sun Y, Ihejirika SA, **Chiang CWK**, Darst BF, Song S, Shen Y, Ye K. "Gene-diet interaction analysis in UK Biobank identified genetic loci that modify the association between fish oil supplementation and the incidence of dementia." **Curr Dev Nutr.** 2025 Aug 5;9(9):107524. doi: 10.1016/j.cdnut.2025.107524.
63. **Tang J, Chiang CWK[†]**. "A genealogy-based approach for revealing ancestry-specific structures in admixed populations." **Am J Hum Genet.** 2025 Aug 7;112(8):1906-1922. doi: 10.1016/j.ajhg.2025.06.016. Epub 2025 Jul 21.
62. **Lo YC, Tian H, Chan TF, Jeon S, Alatorre K, Dinh B**, Maskarinec G, Taparra K, Nakatsuka N, Yu M, Chen CY, Lin YF, Wilkens LR, Le Marchand L, Haiman CA, **Chiang CWK[†]**. "The accuracy of polygenic score models for BMI and Type II diabetes in the Native Hawaiian population." **Commun Biol.** 2025 Apr 23;8(1):651. doi: 10.1038/s42003-025-08050-7.
61. **Fan C[†], Cahoon JL, Dinh BL**, Ortega-Del Vecchio VD, Huber CD, Edge MD, Mancuso N, **Chiang CWK[†]**. "A likelihood-based framework for demographic inference from genealogical trees." **Nat Genet.** 2025 Apr;57(4):865-874. doi: 10.1038/s41588-025-02129-x. Epub 2025 Mar 20.
60. Metayer C, Spector LG, Scheurer ME, **Jeon S**, Scott RJ, Takagi M, Clavel J, Manabe A, Ma X, Hailu EM, Lupo PJ, Milne E, Urayama KY, Bonaventure A, Kato M, Meirhaeghe A, **Chiang CWK**, Morimoto LM, Wiemels JL. "One carbon (folate) metabolism and development of childhood acute lymphoblastic leukemia: a genetic pathway analysis from the Childhood Cancer and Leukemia International Consortium." **Cancer Epidemiol Biomarkers Prev.** 2024 Jun 21. doi: 10.1158/1055-9965.EPI-24-0189. Online ahead of print
59. **Cahoon JL, Rui X*, Tang E*, Simons C, Langie J, Chen M, Lo YC, Chiang CWK[†]**. "Imputation Accuracy Across Global Human Populations." **Am J Hum Genet.** 2024 May 2;111(5):979-989. doi: 10.1016/j.ajhg.2024.03.011. Epub 2024 Apr 10.
58. Hangai M, Kawagushi T, et al. "Genome-wide assessment of genetic risk loci for childhood acute lymphoblastic leukemia in Japanese patients." **Haematologica.** 2024 Apr 1;109(4):1247-1252. doi: 10.3324/haematol.2023.282914.
57. de Smith AJ*, Wahlster L*, **Jeon S***, Kachuri L, Black S, **Langie J**, Cato LD, Nakatsuka N, **Chan TF**, Xia G, Mazumder S, Yang W, Gazal S, Eng C, Hu D, Buchard EG, Ziv E, Metayer C, Mancuso N, Yang JJ, Ma X, Wiemels JL, Yu F*, **Chiang CWK***, Sankaran VG*. "A noncoding regulatory variant in *IKZF1* increases acute lymphoblastic leukemia risk in Hispanic/Latino children." **Cell Genom.** 2024 Apr 10;4(4):100526. doi: 10.1016/j.xgen.2024.100526. Epub 2024 Mar 26.
 - NCI Epidemiology and Genomics Research Program (EGRP) 2024 Research highlights
56. Brandt DYC[†], Huber CD[†], **Chiang CWK[†]**, Ortega-Del Vecchio VD[†]. "The Promise of Inferring the Past using the Ancestral Recombination Graph." **Genome Biol Evol.** 2024 Feb 1;16(2):evae005.

doi: 10.1093/gbe/evae005.

- Recognized as among GBE's most cited recent article: 45 citations as tracked by published as of July 2025

55. **Dinh BL, Tang E**, Taparra K, Nakatsuka N, Chen F, **Chiang CWK[†]**. "Recombination map tailored to Native Hawaiians may improve robustness of genomic scans for positive selection." **Hum Genet.** 2024 Jan;143(1):85-99. doi: 10.1007/s00439-023-02625-2. Epub 2023 Dec 29.
54. Link V, Schraiber JG, **Fan C, Dinh B**, Mancuso N, **Chiang CWK**, Edge MD. "Tree-based QTL mapping with expected local genetic relatedness matrices." **Am J Hum Genet.** 2023 Dec 7;110(12):2077-2091. doi: 10.1016/j.ajhg.2023.10.017.
53. **Chan TF, Rui X**, Conti DV, Fornage M, Graff M, Haessler J, Haiman C, Highland HM, Jung SY, Kenny E, Kooperberg C, Le Marchand L, North KE, Tao R, Wojcik G, Gignoux CR, PAGE Consortium, **Chiang CWK***, Mancuso N*. "Estimating heritability explained by local ancestry and evaluating stratification bias in admixture mapping from summary statistics." **Am J Hum Genet.** 2023 Nov 2;110(11):1853-1862. doi: 10.1016/j.ajhg.2023.09.012. Epub 2023 Oct 23.
52. **Jeon S, Lo YC**, Morimoto LM, Metayer C, Ma X, Wiemels JL, de Smith AJ, **Chiang CWK[†]**. "Evaluating Genomic Polygenic Risk Scores for Childhood Acute Lymphoblastic Leukemia in Latinos." **HGG Adv.** 2023 Sep 13:100239. doi: 10.1016/j.xhgg.2023.100239.
51. **Sheng X**, Xia L, **Cahoon JL**, Conti DV, Haiman CA, Kachuri L, **Chiang CWK[†]**. "Inverted genomic regions between reference genome builds in humans impact imputation accuracy and decrease the power of association testing." **HGG Adv.** 2022 Nov 11;4(1):100159. doi: 10.1016/j.xhgg.2022.100159. eCollection 2023 Jan 12.
50. Li S, **Chiang CWK**, Myint SS, Arroyo K, **Chan TF**, Morimoto L, Metayer C, de Smith AJ, Walsh KM, Wiemels JL. "Localized variation in ancestral admixture identifies pilocytic astrocytoma risk loci among Latino children." **PLoS Genet.** 2022 Sep 7;18(9):e1010388. doi: 10.1371/journal.pgen.1010388.
49. Qin X, **Chiang CWK**, Gaggiotti OE. "Deciphering signatures of natural selection via deep learning." **Brief Bioinform.** 2022 Sep 2:bbac354. doi: 10.1093/bib/bbac354.
48. Yu F, Cato LD, Weng C, Liggett LA, **Jeon S**, Xu K, **Chiang CWK**, Wiemels JL, Weissman JS, de Smith AJ, Sankaran VG. "Variant to function mapping at single-cell resolution through network propagation." **Nat Biotechnol.** 2022 Jun 6. doi: 10.1038/s41587-022-01341-y.
47. Qin X, **Chiang CWK**, Gaggiotti OE. "KLFDAPC: a supervised machine learning approach for spatial genetic structure analysis." **Brief Bioinform.** 2022 Jun 2:bbac202. doi: 10.1093/bib/bbac202.
46. **Fan C[†]**, Mancuso N*, **Chiang CWK*[†]**. "A genealogical estimate of genetic relationships." **Am J Hum Genet.** 2022 May 5;109(5):812-824. doi: 10.1016/j.ajhg.2022.03.016.
45. Fernandez-Rhodes L, Graff M, et al. "Ancestral diversity improves discovery and fine-mapping of genetic loci for anthropometric traits – the Hispanic/Latino Anthropometry Consortium." **HGG Adv.** 2022 Mar 11;3(2):100099. doi: 10.1016/j.xhgg.2022.100099.
44. **Jeon S**, de Smith AJ, Li S, **Chen M, Chan TF**, Muskens IS, Morimoto LM, Dewan AT, Mancuso N, Metayer C, Ma X, Wiemels JL[†], **Chiang CWK[†]**. "Genome-wide trans-ethnic meta-analysis identifies novel susceptibility loci for childhood acute lymphoblastic leukemia." **Leukemia.** 2022 Mar;36(3):865-868. doi: 10.1038/s41375-021-01465-1.
43. Darst BF, Hughley R, et al. "A rare germline *HOXB13* variant contributes to risk of prostate cancer in

men of African ancestry." **Eur Urol**. 2022 Jan 11:S0302-2838(21)02271-5. doi: 10.1016/j.eururo.2021.12.023.

42. **Chen M[†], Chiang CWK[†]**. "Allele frequency differentiation at height-associated SNPs among continental human populations." **Eur J Hum Genet**. 2021 Oct;29(10):1542-1548. doi: 10.1038/s41431-021-00938-2.
41. Kachuri L, **Jeon S**, DeWan AT, Metayer C, Ma X, Witte JS, **Chiang CWK**, Wiemels JL, de Smith AJ. "Genetic determinants of blood cell traits influence susceptibility to childhood acute lymphoblastic leukemia." **Am J Hum Genet**. 2021 Oct 7;108(10):1823-1835. doi: 10.1016/j.ajhg.2021.08.004.
40. **Chiang CWK[†]**. "The opportunities and challenges of integrating population histories into genetic studies for diverse populations: a motivating example from Native Hawaiians." **Front Genet**. 2021 Sep 27;12:643883. doi: 10.3389/fgene.2021.643883.
39. Ganel L, Chen L, Christ R, Vangipurapu J, Young E, Das I, Kanchi K, Larson D, Regier A, Abel H, Kang CJ, Scott A, Havulinna A, **Chiang CWK**, Service S, Freimer N, Palotie A, Ripatti S, Kuusisto J, Boehnke M, Laakso M, Locke A, Stitzel NO, Hall IM. "Mitochondrial genome copy number measured by DNA sequencing in human blood is strongly associated with metabolic traits via cell-type composition differences." **Hum Genomics**. 2021 Jun 7;15(1):34. doi: 10.1186/s40246-021-00335-2.
38. Chen L, Abel HJ, Das I, Larson DE, Ganel L, Kanchi KL, Regier AA, Young EP, Kang CJ, Scott AJ, Chiang C, Wang X, Lu S, Christ R, Service SK, **Chiang CWK**, Havulinna AS, Kuusisto J, Boehnke M, Laakso M, Palotie A, Ripatti S, Freimer NB, Locke AE, Stitzel NO, Hall IM. "Association of structural variation with cardiometabolic traits in Finns." **Am J Hum Genet**. 2021 Apr 1;108(4):583-596. doi: 10.1016/j.ajhg.2021.03.008.
37. Graff M, et al. "Discovery and fine-mapping of height loci via high-density imputation of GWASs in individuals of African ancestry." **Am J Hum Genet**. 2021 Apr 1;108(4):564-582. doi: 10.1016/j.ajhg.2021.02.011.
36. **Sun H*, Lin M***, Russell EM, Minster RL, **Chan TF, Dinh BL**, Naseri T, Reupena MS, Lum-Jones A, OLaGA Study Group, Cheng I, Wilkens LR, Le Marchand L, Haiman CA, **Chiang CWK[†]**. "The impact of global and local Polynesian genetic ancestry on complex traits in Native Hawaiians." **PLoS Genet**. 2021 Feb 11; doi: 10.1371/journal.pgen.1009273.

Contributed Abstracts: Talks (Last 5 years)

bolded name denotes group member.

34. **Chiang CWK, Muthukrishnan E, Dinh BL**, Sheng X, Chen F, Taparra K, Wilkens LR, Le Marchand L, Haiman CA, **Tang J**. "The trans-Pacific voyage that shaped the pattern of genetic variation and enrichment of functional alleles in Native Hawaiians." Society of Molecular Biology and Evolution (SMBE), Copenhagen, Denmark, Jun 2026.
33. **Chiang CWK, Dinh BL, Wang X**, Sheng X, Wan P, Srivastava AK, Naseri T, Viali S, Wilkens LR, Le Marchand L, Haiman CA, Weeks DE, Carlson JC. "Meta-analysis of over 8,000 individuals from Hawai'i and Samoa for genetic associations to cardiometabolic phenotypes." Pasifikas in Medicine, Honolulu, HI, Apr 2026.
32. Brown SC, White K, Kana'iaupuni M, **Chiang CWK**, Wilkens L, Binder AM. "Genetic predisposition to Type 2 Diabetes increases stroke risk across diverse populations." International Stroke Conference (ISC), New Orleans, LA, Feb 2026.

31. **Chiang CWK, Malomane DK**, Williams MP, Huber CD, Mangul S, Abedalthagafi M. "Patterns of population structure and genetic variation within the Saudi Arabian population." American Association of Biological Anthropologists (AABA), Baltimore, MD, Mar 2025.
30. Izarraras-Gomez A, Huber CD, **Chiang CWK**, Ortega-Del Vecchyo VD. "Inference of the distribution of fitness effects using local genealogies." Probabilistic Modeling in Genomics (ProbGen), Cold Spring Harbor, NY, Mar 2025.
29. **Tang J** and **Chiang CWK**. "A genealogy-based approach for revealing ancestry-specific structures in admixed populations." American Society of Human Genetics (ASHG), Denver, CO, Nov 2024.
28. **Chiang CWK** and **Fan C**. "A genealogy-based framework to estimate population structure and demographic history." Society of Molecular Biology and Evolution (SMBE), Puerto Vallarta, Mexico, Jul 2024.
27. **Chiang CWK** and **Fan C**. "A genealogy-based framework to estimate population structure and demographic history." The Allied Genetics Conference (TAGC), Washington DC, Mar 2024.
26. **Cahoon JL**, Mathieson I, **Chiang CWK**, Mathieson S. "Leveraging deep learning to infer archaic human introgression from ancestral recombination graphs." The Allied Genetics Conference (TAGC), Washington DC, Mar 2024. (withdrawn due to inability to attend)
25. **Chiang CWK**. "A genealogy-based framework to estimate population structure and demographic history." The 3rd AsiaEvo Conference, Singapore, Dec 2023.
24. **Chiang CWK, Cahoon JL, Rui X, Tang E, Simons C, Langie J, Chen M, Lo YC**. "Imputation Efficacy Across Diverse Global Human Populations." International Genetic Epidemiology Society (IGES), Nashville, TN, Nov 2023. Poster Highlight Talk / Best Poster Award
23. **Langie J, Jeon S**, Ma X, Metayer C, de Smith AJ, Wiemels JL, **Chiang CWK**. "Identifying Acute Lymphoblastic Leukemia Risk Loci in Latino Children via Admixture Mapping." American Society of Human Genetics (ASHG), Washington DC, Nov 2023.
22. **Chiang CWK, Cahoon JL, Rui X, Tang E, Simons C, Langie J, Chen M, Lo YC**. "Imputation Efficacy Across Diverse Global Human Populations." International Society for Evolution, Medicine & Public Health (ISEMPH), Irvine, CA, Aug 2023.
21. **Cahoon JL, Rui X, Tang E, Simons C, Langie J, Chen M, Lo YC, Chiang CWK**. "Imputation Efficacy Across Global Human Populations." Southern California Evolutionary Genetics and Genomics Meeting (Scale), Irvine, CA, May 2023.
20. **Dinh BL, Tang E**, Wilkens LR, Le Marchand L, Haiman CA, **Chiang CWK**. "Population-specific Recombination Maps Reduce False Positive Findings in Genomic Scans of Adaptations." Southern California Evolutionary Genetics and Genomics Meeting (Scale), Irvine, CA, May 2023.
19. **Jeon S**, Gazal S, Mancuso N, Meyayer C, Ma X, Wiemels JL, **Chiang CWK**, de Smith AJ. "Risk allele associated with childhood acute lymphoblastic leukemia at the *IKZF1* locus is associated with Indigenous American ancestry and absent in European-ancestry populations." American Society of Human Genetics (ASHG), Los Angeles, CA, Oct 2022.
18. Link V, **Fan C, Chiang CWK**, Mancuso N, Edge MD. "ARG-based association mapping." Society of Molecular Biology and Evolution (SMBE) Everywhere Symposium, virtual, Sep 2022.

17. **Fan C**, Mancuso N, **Chiang CWK**. “A likelihood-based framework for demographic inference from genealogical trees.” Society of Molecular Biology and Evolution (SMBE) Everywhere Symposium, virtual, Sep 2022.
16. Link V, **Fan C**, **Chiang CWK**, Mancuso N, Edge MD. “ARG-based association mapping.” Population, Evolutionary, and Quantitative Genetics (PEQG), Pacific Grove, CA, Jun 2022.
15. Hughley R, Darst BF, Andrews C, Berndt SI, Sheng X, Kote-Jaral Z, Lachance J, **Chiang CWK**, Jalloh M, Abediyi A, Aisuodionoe-Shadrac O, Adusei B, Mante S, Mensah J, Adjei A, Chanock SJ, Watya S, Eeles RA, Rebbeck TR, Conti DV, Haiman CA, MADCaP Consortium, PRACTICAL Consortium, and African Ancestry Prostate Cancer Consortium. “Germline *HOXB13* variant contributes to risk of prostate cancer in men of African ancestry.” American Society of Human Genetics (ASHG), virtual, Oct 2021.
14. **Chiang CWK**, **Fan C**, Mancuso N. “A genealogical estimate of genetic relationships to improve detection of population structure over time.” International Genetic Epidemiology Society (IGES), virtual, Oct 2021.

Contributed Abstracts: Posters (Last 2 Years)

From 2016-2026, 13 of 38 submitted abstracts at ASHG received the Reviewer's Choice Award or Platform Talk. **bolded** name denotes group member.

85. **Tang J**, **Muthukrishnan E**, **Dinh BL**, Sheng X, Chen F, Taparra K, Wilkens LR, Le Marchand L, Haiman CA, **Chiang CWK**. “The trans-Pacific voyage that shaped the pattern of genetic variation and enrichment of functional alleles in Native Hawaiians.” American Society of Human Genetics (ASHG), Montreal, Canada, Oct 2026. *Reviewer's Choice Abstract*
84. Xie J, Tan Q, Synder LA, Chen F, Trost B, Fernandez B, **Chiang CWK**, Mancuso N, Shu C. “Rare CNV burden and heritability of machine-learning-derived autism subtypes.” American Society of Human Genetics (ASHG), Montreal, Canada, Oct 2026.
83. Neuffer TL, Darroch H, Calhoun C, Takei R, Pascart T, **Wang X**, Gazal S, **Chiang CWK**, Yerges-Armstrong L, Castel SE, Merriman TR, Leask MP. “Characterising the role of *opfl1* in the inflammatory response in zebrafish.” Queenstown Research Week, Dunedin, New Zealand, Aug 2026.
82. Morice E, Takei R, Pascart T, **Wang X**, Gazal S, **Chiang CWK**, Yerges-Armstrong L, Castel SE, Ledgerwood EC, Merriman TR, Leask MP. “From GWAS to mechanism: developing a BlaER1-based framework for functional investigation of gout risk loci.” Queenstown Research Week, Dunedin, New Zealand, Aug 2026.
81. **Lee EY**, **Dinh BL**, **Tang J**, Streicher SA, **Wang X**, **Biswas S**, **Tian H**, Yu X, Taparra K, Naseri T, Viali S, Weeks DE, Carlson JC, Haiman CA, Le Marchand L, Maskarinec G, Wilkens LR, Park SY, **Chiang CWK**. “Data-driven dietary score complement polygenic prediction models for BMI in trans-ethnic cohorts.” American Society of Nutrition (ASN), National Harbor, MD, Jul 2026.
80. Streicher SA, Park SY, **Chiang CWK**, Sheng X, Filowski MK, Conti DV, Haiman CA, Wilkens LR, Le Marchand L. “Developing the traditional Okinawan diet score in the Multiethnic Cohort Study.” American Society of Nutrition (ASN), National Harbor, MD, Jul 2026.
79. Izarraras-Gomez A, Huber CD, Chiang CWK, Ortega-Del Vecchyo VD. “Inference of the distribution of fitness effects using local genealogies.” Society of Molecular Biology and Evolution (SMBE), Copenhagen, Denmark, Jun 2026.

78. **Tian L, Malomane DK, Tang J, Abedalthagafi A, Chiang CWK.** “Re-interpreting deleterious variant burden in populations with elevated rates of runs of homozygosity.” Population, Evolutionary, and Quantitative Genetics (PEQG), Pacific Grove, CA, Jun 2026.
77. **Tang J, Muthukrishnan E, Dinh BL,** Sheng X, Chen F, Tappara K, Wilkens LR, Le Marchand L, Haiman CA, **Chiang CWK.** “The trans-Pacific voyage that shaped the pattern of genetic variation and enrichment of functional alleles in Native Hawaiians.” Population, Evolutionary, and Quantitative Genetics (PEQG), Pacific Grove, CA, Jun 2026.
76. **Muthukrishnan E, Tang J, Chiang CWK.** “Differentiating shared ancient origin and recent admixture for genetic similarities between populations.” Population, Evolutionary, and Quantitative Genetics (PEQG), Pacific Grove, CA, Jun 2026.
75. **Tian L, Malomane DK, Tang J, Abedalthagafi M, Chiang CWK.** “Re-interpreting Deleterious Variant Burden in the Highly Consanguineous Saudi Arabian Population.” Southern California Evolutionary Genetics and Genomics (SCalE) Meeting, Los Angeles, CA, Nov 2025.
74. Tan Q, Xie J, Synder LG, Fernandez B, Mancuso N, **Chiang CWK,** Shu C. “Machine learning-derived subtypes of autism reveal distinct genetic contributions from common and rare variants.” World Congress of Psychiatric Genetics (WCPG), Cancún, Mexico, Oct 2025.
73. **Dinh BL, Wang X, Alatorre K, Tian H, Tang J, Lee EY,** Sheng X, Wilkens LR, Le Marchand L, Haiman CA, Weeks DE, Carlson JC, **Chiang CWK.** “Genetic genealogy-based approach enhances the discovery of trait associated loci in small-sample populations.” American Society of Human Genetics (ASHG), Boston, MA, Oct 2025.
72. **Lee EY, Dinh BL, Tang J,** Streicher S, **Biswas S, Tian H,** Tappara K, Haiman CA, Le Marchand L, Maskarinec G, Wilkens LR, Park SY, **Chiang CWK.** “Incorporating dietary information to enhance polygenic prediction models with applications to body mass index and type-2 diabetes.” American Society of Human Genetics (ASHG), Boston, MA, Oct 2025.
71. **Tang J, Dinh BL,** Sheng X, Chen F, Tappara K, Wilkens LR, Le Marchand L, Haiman CA, **Chiang CWK.** “Pattern of genetic variation and enrichment of functional alleles in Native Hawaiians.” American Society of Human Genetics (ASHG), Boston, MA, Oct 2025.
70. **Alatorre K, Dinh BL, Tian H,** Sheng X, **Lee EY, Tang J,** Tappara K, Le Marchand L, Wilkens LR, Haiman CA, **Chiang CWK.** “An evaluation of the prediction accuracy of polygenic score models across thirteen quantitative biomarkers and anthropometric measures in Native Hawaiians.” American Society of Human Genetics (ASHG), Boston, MA, Oct 2025.
69. **Tian H, Chan, TF,** Sheng X, Wan P, **Dinh BL, Tang J, Langie J,** Bogumil D, Tappara K, Wilkens LR, Le Marchand L, Haiman CA, Conti DV, Chiang CWK. “The predictive performance of cancer polygenic score models in Native Hawaiians.” American Society of Human Genetics (ASHG), Boston, MA, Oct 2025. Withdrawn due to inability to attend
68. Fuller H, Xu H, Agasaro O, Guo B, Sheng X, Wan P, Wilkens LR, Le Marchand L, Albanes D, Cheng I, Conti DV, Haiman CA, **Chiang CWK,** Ye K, Darst BF. “Factors contributing to circulating metabolite variation in populations with high genetic admixture.” American Society of Human Genetics (ASHG), Boston, MA, Oct 2025.
67. Tan Q, Xie J, Synder LG, Fernandez B, Mancuso N, **Chiang CWK,** Shu C. “Machine learning-derived subtypes of autism reveal distinct genetic contributions from common and rare variants.” American Society of Human Genetics (ASHG), Boston, MA, Oct 2025. Reviewer’s Choice Abstract

66. Morice E, Eboda O, Takei R, Pascart T, **Wang X**, Gazal S, **Chiang CWK**, Yerges-Armstrong L, Castel SE, Merriman TR, Leask MP. “Understanding the role of genetic variation in immune biology in Polynesian populations.” Queenstown Research Week (QRW), Christchurch, New Zealand, Aug 2025.
65. **Chiang CWK**, Edge MD, Link V, **Tang J**, **Dinh BL**, **Cahoon J**, Schraiber JG, Ortega-Del Vecchyo VD, Huber CD, Mancuso N, **Fan C**. “A genealogy-based framework to infer the demographic history, genetic structure, and phenotype association.” Probabilistic Modeling in Genomics (ProbGen), Cold Spring Harbor, NY, Mar 2025.
64. Liu T, **Langie J**, Yang W, Morimoto LM, Ma X, Metayer C, Lupo PJ, Scheurer ME, Yang JJ, Wiemels JL, **Chiang CWK**, de Smith AJ. “Genome-wide association study of acute lymphoblastic leukemia in Hispanic/Latino children identifies a putatively novel risk locus at chr5q31.1.” American Society of Hematology (ASH), San Diego, CA, Dec 2024.
63. **Dinh BL**, **Wang X**, **Tian H**, **Tang J**, Sheng X, Carlson JC, Weeks DE, Wilkens LR, Le Marchand L, Haiman CA, **Chiang CWK**. “Statistical and population genetic approaches to enhance the discovery of trait-associated loci in understudied populations with small sample size.” American Society of Human Genetics (ASHG), Denver, CO, Nov 2024. *Reviewer’s Choice Abstract*
62. **Wang X**, **Dinh BL**, Sheng X, Young E, Leask M, Taparra K, Stamp LK, Dalbeth N, Murphy R, de Zoysa J, Naseri T, Reupena MS, Viali S, Castel SE, Merriman TR, Carlson JC, Weeks DE, Wilkens LR, Le Marchand L, Hall I, Haiman CA, Stitzel N, **Chiang CWK**. “A multi-ancestry reference panel from 10,721 individuals to improve the genotype imputation for individuals of East Asian- and Polynesian-ancestry.” American Society of Human Genetics (ASHG), Denver, CO, Nov 2024.

Invited Talks and Colloquia

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| 2026 | Division of Nephrology & Hypertension, University of Utah, Salt Lake City, UT |
| 2025 | Institute of Epidemiology and Preventive Medicine, National Taiwan University, Taipei, Taiwan |
| 2025 | Institute of Biomedical Sciences (IBMS), Academia Sinica, Taipei, Taiwan |
| 2025 | School of Life Sciences, Fudan University, Shanghai, China |
| 2025 | Center for Interdisciplinary Theoretical and Mathematical Sciences, RIKEN (The Institute of Physical and Chemical Research), Tokyo, Japan |
| 2025 | Department of Genome Informatics, University of Tokyo, Tokyo, Japan |
| 2025 | Population Sciences in the Pacific Program, University of Hawai’i Cancer Center, Honolulu, HI |
| 2025 | Division of Nephrology & Hypertension, University of Utah, Salt Lake City, UT |
| 2025 | Department of Biostatistics, University of Michigan, Ann Arbor, MI |
| 2024 | Department of Biomolecular Engineering, University of California, Santa Cruz, CA |
| 2024 | Institute for Genomic Health, Icahn School of Medicine at Mount Sinai, New York, NY |
| 2023 | Institute for Genome Sciences, Department of Life Sciences, National Yang Ming Chiao Tung University, Taipei, Taiwan |
| 2023 | Institute of Epidemiology and Preventive Medicine, National Taiwan University, Taipei, Taiwan |
| 2023 | Department of Genetics, University of Pennsylvania, Philadelphia, PA |
| 2023 | Genomic Medicine Institute Seminar, Department of Biomedical Sciences, Seoul National University College of Medicine, Seoul, Korea |
| 2023 | Department of Molecular and Human Genetics, Baylor College of Medicine, Houston, TX |
| 2022 | Population Biology Seminar Series, University of Southern California, Los Angeles, CA |
| 2022 | Gerardo Heiss Cardiovascular Epidemiology Seminar, University of North Carolina at Chapel |

	Hill, Chapel Hill, NC
2022	Native Hawaiian Community Advisory Board (outreach), University of Hawai'i Cancer Center, Honolulu, HI
2021	American Association for Cancer Research (AACR) Conference on The Science of Cancer Health Disparity, Session on "Heterogeneity Within the Asian and Pacific Islander Diaspora"
2020	Department of Genetics, University of Georgia, Athens, GA
2019	International Laboratory of Human Genome Research (LIIGH), National University of Mexico (UNAM), Queretero, Mexico
2019	Quantitative Computational Biology Section, Department of Biological Sciences, University of Southern California, Los Angeles, CA
2018	Informatics Center for Neurogenetics and Neurogenomics (ICNN) Symposium, University of California, Los Angeles (UCLA), Los Angeles, CA
2018	Asian Evo Conference, Symposium on "Human Evolution and Adaptation", China National GeneBank, Shenzhen, China
2018	Institute of Biomedical Sciences (IBMS), Academia Sinica, Taipei, Taiwan
2017	Center for Genetic Epidemiology, Department of Preventive Medicine, University of Southern California, Los Angeles, CA
2017	Virginia Institute for Psychiatric and Behavioral Genetics, Virginia Commonwealth University, Richmond, VA
2017	Department of Anthropology, University of Utah, Salt Lake City, UT
2015	CAS-MPG Partnered Institute of Computational Biology, Chinese Academy of Science, Shanghai, China
2015	Center for Genomic Sciences, University of Hong Kong, Hong Kong, SAR, China
2015	Institute of Biomedical Sciences (IBMS), Academia Sinica, Taipei, Taiwan

Academic Awards and Recognitions

2024	Dean's Early Career Award for Excellence in Mentoring, Keck School of Medicine, USC
2023	Human Genetics and Genomics Advances Early Career Investigator Award
2020	Undergraduate Research Associate Program Award, USC (To support undergraduate researchers S. Rashid and S. Sommerer in AY 2020-21)
2017	ASHG Charles J. Epstein Trainee Award for Excellence in Human Genetics Research (Semifinalist)
2013-2015	National Institute of Health NRSA Postdoctoral Fellowship, F32
2010	Summer Institute in Statistical Genetics Travel Scholarship
2007-2010	National Science Foundation Graduate Fellowship
2005	Director's Fellowship, Yale University (declined)
2005	Phi Beta Kappa Society
2004	Ira J. and Shirley Spoon Honors Collegium Scholarship, UCLA
2003	Golden Key International Honour Society
2001	CRC Press Freshmen Chemistry Achievement Award
2001	The National Society of Collegiate Scholar
2001	Alpha Lambda Delta Honor Society
2001	Phi Eta Sigma Honor Society

Mentored Student, Staff, and Postdocs

Abbreviations: **PIBBS**, Programs in Biomedical and Biological Sciences PhD program; **CBB**, Computational Biology and Bioinformatics PhD program; **MBBO**, Marine Biology and Biological Oceanography PhD program; **QB**, Quantitative Biology undergraduate program, USC; **HB**, Human Biology undergraduate program, USC; **CS**, Computer Science undergraduate program, USC.

Year	Name	Program or Position	Role
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Current Trainee**Postdocs**

2023-current	Ji Tang, Ph.D.	Postdoctoral Fellow, USC	Postdoc advisor
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Students

2020-current	Bryan Dinh	Ph.D. student, CBB, USC	Ph.D. advisor
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2021-current	Jalen Langie	Ph.D. student, Epidemiology, USC NIH F31 Predoc Fellowship (2023-26)	Ph.D. advisor
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2023-current	He Tian	Ph.D. student, Epidemiology, USC	Ph.D. advisor
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2024-current	Xinran Wang	Ph.D. student, Epidemiology, USC	Ph.D. advisor
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2025-current	Leqi Tian	Ph.D. student, PIBBS, USC	Ph.D. advisor
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2025-current	Lan Trieu	B.S. student, QB, USC	--
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2026-current	Junhyuk Jay Lee	B.S. / M.S. student, QB, USC	--
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Research Staff

2023-current	Eunice Lee, Ph.D.	Senior Research Associate, USC	--
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2024-current	Eaaswarkhanth Muthukrishnan, Ph.D.	Research Associate, USC	--
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2024-current	Kimberli Alatorre	Programmer Analyst I, USC	--
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2025-current	Subarna Biswas, M.D. Ph.D.	Research Scientist, USC	--
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2026-current	Jonathan Mah, Ph.D.	Research Associate, USC	--
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Graduate Student Committees

2019-2023	Olzhas Iksan	Ph.D. student, Genetics, Kazakh National University	Ph.D. Dissertation (external member)
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2020	Maria Ruggeri	Ph.D. student, MBBO, USC	Ph.D. Qualifying (external member)
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2021-2022	Qianxi Feng	Ph.D. student, Epidemiology, USC	Ph.D. Dissertation
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2021	Changqing Su	M.S. student, Biostatistics, USC	M.S. Thesis
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2021-2022	Shaobo Li	Ph.D. student, PIBBS, USC	Ph.D. Dissertation
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2022-2023	Yating Zeng	M.S. student, Biostatistics, USC	M.S. Thesis
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2022-current	Huifang Xu	Ph.D. student, Genetics, University of Georgia	Ph.D. Dissertation (external member)
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2023	Michaela Ince	Ph.D. student, PIBBS, USC	Ph.D. Qualifying
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2023	Bida Gu	Ph.D. student, CBB, USC	Ph.D. Qualifying
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2024-current	Lu Wang	Ph.D. student, Genetics, University of Georgia	Ph.D. Dissertation (external member)
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2025	Janis (Junjian) Liu	Ph.D. student, CBB, USC	Ph.D. Qualifying
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2025	Walfred (Wangfei) Ma	Ph.D. student, CBB, USC	Ph.D. Qualifying
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2025	Vignesh Arunachalam	Ph.D. student, Biomedical Sciences Queensland Univ. of Technology	Ph.D. Dissertation (external reviewer)
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2026	Yunlu Li	M.S. student, Applied Biostatistics and Epidemiology	M.S. Thesis
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Former Trainee

2018-2019	Hanxiao Sun	M.S. student, Biostatistics, USC	MS thesis advisor
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2018-2020	Meng Lin, Ph.D.	Postdoctoral Fellow, USC	Postdoc advisor
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2018-2021	Minhui Chen, Ph.D.	Postdoctoral Fellow, USC	Postdoc advisor
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2018-2023	Soyoung Jeon	Ph.D. student, PIBBS, USC	Ph.D. advisor
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		Postdoctoral Fellow, USC	Postdoc advisor
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2019-2020	Samuel Sommerer	B.S. student, QB/CS, USC	--
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2019-2023	Caoqi Fan	Ph.D. student, CBB, USC	Ph.D. advisor
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2019-2025	Tsz Fung Chan	Ph.D. student, Epidemiology, USC	Ph.D. co-advisor
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2020-2021	Sydney Rashid	B.S. student, QB, USC	--
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2020-2022	Xinyue Rui	M.S. student, Biostatistics, USC	MS thesis advisor
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		B.S. student, Mathematics, USC USC Provost Fellowship (2020, 21) Jennifer Battat USC Scholarship	
2021-2023	Ying-Chu Lo, Ph.D.	Postdoctoral Fellow, USC USC-Taiwan Postdoc Fellowship	Postdoc advisor
2021-2023	Christopher Simons	B.S. / M.S. student, QB, USC	--
2021-2023	Echo Tang	B.S. student, QB, USC	--
2021-2024	Jordan Cahoon	B.S. student, CS, USC USC Viterbi Fellowship (2020-24) Barry Goldwater Scholarship (2023) Discovery Fellowship (2024) QCB Google Alumni Award (2024)	--
2022-2025	Dorcus Malomane, Ph.D.	Postdoctoral Fellow, USC	Postdoc advisor
2023	Melody McBride	B.S. student, HB, USC	--
2023-2024	Xinran Wang	Programmer Analyst I, USC	--
2023-2025	Indu Sharma, Ph.D.	Postdoctoral Fellow, USC	Postdoc advisor
2024-2025	Rishee Venkatesh	B.S. student, QB, USC	--
2025-2026	Ke Wu	M.S. student, Transl. Biotech., USC	--

Rotation and Other Visiting Students

2018 Spr	Kuochang Tseng	Ph.D. Rotation, PIBBS, USC	--
2018 Spr	Charles Bramlett	Ph.D. Rotation, PIBBS, USC	--
2018 Fall	Audrey Nickle	Ph.D. Rotation, PIBBS, USC	--
2021 Sum	Dandan Peng	Ph.D. Rotation, CBB, USC	--
2022 Spr	Ilkwon Cho	Ph.D. Rotation, CBB, USC	--
2023 Sum	Zachary Chang	High school intern	--
2023 Fall	Yinqiao Wang	Ph.D. Rotation, Epidemiology, USC	--
2024 Spr	Yifan Zhang	Ph.D. Rotation, Epidemiology, USC	--
2024 Spr	Juanita Quino	Ph.D. Rotation, Epidemiology, USC	--
2024 Spr	Matthew Williams	Visiting postdoc, Penn State U	--
2024 Sum	Evelyn Kim	High school intern	--
2024 Sum	Kimberli Alatorre	Post-baccalaureate intern	--
2025 Spr	Jingyu Xie	Ph.D. Rotation, Epidemiology, USC	--

Teaching

2021-present	Course Director or co-Director (since 2024), University of Southern California PM 534 (Statistical Genetics: Introduction to Medical Population Genetics) 3 units, Fall Term
2025-2026	Guest Lecturer, University of Utah Haumana 'O Pasifika Summer Research Program, 1hr/term, Summer Term
2020	Guest Lecturer, University of Southern California PM 533 (Genetic and Molecular Epidemiology), 1hr/term, Fall Term
2020	Guest Lecturer, University of Hawai'i Cancer Center Multiethnic Cohort T32 Postdoctoral Fellowship Seminar Series, 1hr, Spring Term
2019-2021	Guest Lecturer, University of Southern California BISC 577A (Computational Biology Laboratory), 2hrs/term, Fall Term
2009	Teaching Assistant, Harvard University. LS1B (Genetics, Genomics, and Evolution), Spring Term
2008	Guest Lecturer, Harvard Medical School. Genetics 228 (Genetics in Medicine; "Using Genetics Databases"), Fall Term
2007	Teaching Assistant, Harvard Medical School. Genetics 228 (Genetics in Medicine), Fall Term

Departmental and University Service

Abbreviations: **CGE**, Center for Genetic Epidemiology; **DPPHS**, Department of Population and Public Health Sciences, USC; **QCB**, Department of Quantitative Computational Biology, USC; **KSOM**, Keck School of Medicine; **PIBBS**, Programs in Biomedical and Biological Sciences PhD program; **CBB**, Computational Biology and Bioinformatics PhD program.

Year	Role	Service
2026	Member	QCB Faculty Fifth Year Review Committee
2025	Reviewer	American Cancer Society pilot grants, DPPHS
2025	Reviewer	Zumberge Award Program
		Research Initiatives and Infrastructure, USC
2025	Panelist	Mastering the Art of Mentorship: A Path to Career Success in Academic Medicine, KSOM
2024	Panelist	Grand Round: Genomic Screening and Preventive Health by Noura Abul-Husn, DPPHS
2023	Member	DPPHS IT/Computing Committee
2021, 2024	Ad hoc Interviewer	MD/PHD Admission Committee
2021	Member	CGE Scientific Review Committee
2021-2022	Member	DPPHS Junior Faculty Committee
2021	Member	DPPHS Statistical Genetics Curriculum Committee
2020-present	Chair	CGE Seminar Organizing Committee
2020-present	Webmaster	CGE webpage design and launch
2020, 2022, 2023, 2025	Ad hoc Interviewer	CBB Ph.D. Admission Committee
2019-present	Member, Co-chair	DPPHS Epidemiology Ph.D. Admission Committee
2019	Member	QCB Faculty Search Committee
2018-2020	Member, ad hoc interviewer	PIBBS Ph.D. Admission Committee
2018	Member	DPPHS Biostatistics Ph.D. Admission Committee

Professional Services and Scientific Communities

Editorship and Editorial Board:

Guest Associate Editor (PLoS Genetics, 2023)

Associate Editor (Frontier in Genetics, 2021-2024)

Review Editor (Frontier in Genetics, 2018-2021)

Invited Manuscript Reviewer:

Nature	Nature Genetics	Nature Communications
Science	Science Advances	PNAS
Cell	Cell Genomics	Molecular Biology and Evolution
Am J Hum Genet	HGG Advances	npj Genomic Medicine
Genetics	G3: Genes, Genomes, Genetics	Human Molecular Genetics
Genome Medicine	Genome Biology	Human Genetics
PLoS Genetics	PLoS Computational Biology	PLoS ONE
Eur J Hum Genet	Bioinformatics	iScience
Human Genomics	Evol Med Public Health	Genom, Proteom & Bioinformatics
BMC Genomics	Communications Biology	Scientific Reports
Frontiers in Genetics	Nutrition and Diabetes	

American Society of Human Genetics (ASHG):

Member, 2007-present

Judge, DNA Day Essay Contest, 2014, 2015, 2017, 2021

Program Abstract Reviewer, Evolutionary and Population Genetics, 2017, 2025

Mentor, Trainee-Mentor Luncheon, Evolution and Population Genetics, 2019

Member, Awards Committee, 2021-2023
Session Moderator, “Novel Aspects of Modeling Genetic Architectures of Complex Traits,” 2024
Session Moderator, “Evolution and Population Genetics,” 2025
Member, Program Committee, 2026-2028

Grant Reviewer (*ad hoc*):

Human Frontier Science Program (Reviewer, 2018)
Medical Research Council NIRG (Reviewer, 2019)
NSF CAREER Award (Reviewer, 2021)
German Israeli Foundation (GIF) for Scientific Research and Development (Reviewer, 2022)
Wellcome Discovery Award (Reviewer, 2024)
NIH Genetic Variation and Evolution (GVE) study section (Reviewer, 2024, 2026)
Israel Science Foundation (ISF) (Reviewer, 2026)
NIH Special Emphasis Panel for MIRA, ZRG1 MGG-D (Reviewer, 2026)

Symposia Organizer:

1st AsiaEvo Conference, “Human Evolution and Adaptation” (2018)
Society of Molecular Biology and Evolution, “Using Ancestral Recombination Graphs (ARGs) to infer Evolutionary Processes” (2022)

Membership:

American Society for Human Genetics (ASHG)
Society for Molecular Biology & Evolution (SMBE)
International Society of Evolution Medicine and Public Health (ISEMPH)
American Association of Physical Anthropologists (AAPA)
International Genetic Epidemiology Society (IGES)
Genetic Society of America (GSA)
American Association for the Advancement of Science (AAAS)
American Association of Biological Anthropologists (AABA)

Media Appearances, Engagement, and Coverage

“Long-read sequencing unearths overlooked autism-linked variants”
The Transmitter, Sep 18, 2025
<https://www.thetransmitter.org/spectrum/long-read-sequencing-unearths-overlooked-autism-linked-variants/>

“Inside AJHG: A Chat with Ji Tang”
American Journal of Human Genetics, *ASHG News*, Aug 4, 2025
<https://www.ashg.org/ajhg/inside-ajhg-a-chat-with-ji-tang/>

“Researchers sequence complex parts of human genome, expanding the future of precision medicine”
STAT News, Jul 23, 2025
<https://www.statnews.com/2025/07/23/landmark-study-presents-most-complete-human-genome-sequence/>

“Filling in genomic blanks for disease studies works better for some groups than others”
Keck School of Medicine News, Apr 10, 2024
<https://keck.usc.edu/news/filling-in-genomic-blanks-for-disease-studies-works-better-for-some-groups-than-others/>

“*IKZF1* Genetic Variant May Contribute to Disparities in Risk of ALL in Pediatric Hispanic and Latino Patients”

The ASCO (American Society of Clinical Oncology) Post, Apr 2, 2024

https://ascopost.com/news/march-2024/ikzf1-genetic-variant-may-contribute-to-disparities-in-risk-of-all-in-pediatric-hispanic-and-latino-patients/#new_tab

“USC researchers find genetic variant contributing to disparities in childhood leukemia risk”

Keck School of Medicine News, Mar 26, 2024

<https://keck.usc.edu/news/usc-researchers-find-genetic-variant-contributing-to-disparities-in-childhood-leukemia-risk/>

“Inside HGG Advances: A Chat with Charleston Chiang”

Human Genetics and Genomics Advances, Jan 20, 2023

<https://www.ashg.org/hgga/inside-hgga-with-charleston-chiang/>

“COVID-19 vaccine eligibility expands to anyone over the age of 16 in L.A. county”

USC Annenberg Media, Apr 13, 2021

<https://www.uscannenbergmedia.com/2021/04/13/covid-19-vaccine-eligibility-expands-to-anyone-over-the-age-of-16-in-la-county/>

“Polynesian ancestry linked to obesity in study of Native Hawaiians”

Genetic Obesity News, Feb 15, 2021

<https://geneticobesitynews.com/2021/02/15/polynesian-ancestry-linked-obesity-diabetes-heart-disease-risk-study/>

“Native Hawaiian genetic ancestry linked to metabolic diseases”

The Academic Times, Feb 11, 2021

<https://academictimes.com/native-hawaiian-genetic-ancestry-linked-to-metabolic-diseases/>

“Global, Local Polynesian Ancestry Influence Disease Risk Among Native Hawaiians”

Genomeweb, Feb 11, 2021

<https://www.genomeweb.com/genetic-research/global-local-polynesian-ancestry-influence-disease-risk-among-native-hawaiians#.YCWpKBNKjBI>

“Genetic ancestry linked to diabetes, heart failure and obesity among Native Hawaiians”

Keck School of Medicine News, Feb 11, 2021

<https://keck.usc.edu/genetic-ancestry-linked-to-diabetes-heart-failure-and-obesity-among-native-hawaiians/>

“Polynesian ancestry linked to obesity, diabetes in Native Hawaiians”

Honolulu Star Advertiser, Feb 11, 2021

<https://www.staradvertiser.com/2021/02/11/breaking-news/polynesian-ancestry-linked-to-obesity-diabetes-in-native-hawaiians/>

“Genetics contributes to diabetes, HF, obesity risk in Native Hawaiian individuals”

Healio, Feb 11, 2021

https://www.healio.com/news/cardiology/20210211/genetics-contributes-to-diabetes-hf-obesity-risk-in-native-hawaiian-individuals?utm_medium=social&utm_source=twitter&utm_campaign=scheduledpost

“Accuracy of Genotyping Chips Called into Question”

The Scientist, Jul 24, 2019

<https://www.the-scientist.com/news-opinion/accuracy-of-genotyping-chips-called-into-question--66198>

“Hidden Stories of Chinese Migration and Culture Found in Giant Genetic Study”

The New York Times, Oct 5, 2018

<https://www.nytimes.com/2018/10/05/science/china-genetics.html>

Outreach, Community Engagements and Services

Haumana ‘O Pasifika Program, University of Utah, Salt Lake City, UT

Summer research program to encourage and enable research and career development opportunities for undergraduate and post-baccalaureate students interested in improving the health of Native Hawaiians and other Pacific Islanders

2025-2026

Guest lecturer, faculty panelist

2026

Samoan Heritage Festival and recruitment for Utah Pacific Islander Diabetes Study

Mālama Makali‘i Community Workday, Kailua, HI

Community work restoring native forests at the Ala Mahamoe site in Moanalua

2026

Participant / Volunteer

‘Aha Ho‘olōkahi, Honolulu, HI

Hosted by Papa Ola Lōkahi, the summit is designed to focus and amplify Native Hawaiian traditional and emerging knowledge/resources around health, healing, and well-being.

2025

Participant

Joint Community Advisory Board Retreat, University of Hawai‘i Cancer Center, Honolulu, HI

First retreat of members of all community advisory boards at UHCC to promote interactions between community members and scientists.

2025

Participant

Hoyt Art Gallery, USC, Los Angeles, CA

Community outreach pairing local visual artist with biomedical researcher to collaborate for innovative artistic representation of ongoing research at Keck School of Medicine

2018, 2022, 2023

Partnered Biomedical Researcher

“A PhD is Not Enough” book club seminar series, Graduate Society in Biostatistics and Epidemiology (GSBE), Department of Population and Public Health Sciences, USC, Los Angeles, CA

2022

Faculty Panelist

“How to get a postdoc” workshop, Dept. of Ecology & Evolutionary Biology, UCLA, Westwood, CA

2015

Panelist

Science in the News, Harvard Medical School, Boston, MA

Science in the News is a free evening seminar series for the public aimed at explaining and discussing today's hottest scientific topics without the hype or exaggeration often introduced by media coverage.

2005-2006

Lecture Coordinator

“Stem Cells: Hope or Hype?” lecture series (2005)

“Obesity and Diabetes” lecture series (2006)

MATHCOUNTS, Suzanne Middle School, Walnut, CA

MATHCOUNTS is a nation-wide program for 6th to 8th grade students designed to elicit interest in mathematics through teamwork and competition.

2003-2005

Head Coach, Suzanne Middle School

1999-2003

Assistant Coach, Suzanne Middle School

2006, 2008, 2013

Volunteer, East San Gabriel Valley Chapter, CA